Polynomial Chaos and Collocation Methods and Their Range of Applicability

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Problem Setup

Governing Equations

Let $\mathscr{D} \subset \mathbb{R}^d$ (typically $d = 3$ in physical space) be a fixed domain with a boundary $\partial\mathscr{D}$ and $\mathbf{x} \in \mathscr{D}$ the physical coordinates. Let $(\Omega, \mathscr{A}, \mathscr{P})$ be a probability space where Ω is the abstract set of elementary events, $\mathscr{A}$ is a σ-algebra of subsets of Ω, and $\mathscr{P}$ is a probability measure on $\mathscr{A}$. Our aim is to approximate the random field $u(\mathbf{x};\boldsymbol{\xi}) : \mathscr{D} \times \Gamma \to \mathbb{R}^m$ satisfying the parameterized partial differential equations:

$$\begin{aligned} \mathscr{L}(\mathbf{x},\boldsymbol{\xi};u) &= 0 \quad \text{in } \mathscr{D}, \\ \mathscr{B}(\mathbf{x},\boldsymbol{\xi};u) &= 0 \quad \text{on } \partial\mathscr{D}, \end{aligned} \tag{1}$$

where $\mathscr{L}$ is a linear or nonlinear differential operator, and $\mathscr{B}$ is a boundary operator. Here $\boldsymbol{\xi}(\omega) = (\xi_1(\omega), \xi_2(\omega), \dots, \xi_N(\omega)) : \Omega \to \Gamma \subseteq \mathbb{R}^N$ is a vector of random parameters defined on $(\Omega, \mathscr{A}, \mathscr{P})$ with probability distribution $\mathscr{P}_{\boldsymbol{\Xi}}$, of which components $\xi_1, \xi_2, \dots, \xi_N$ are mutually independent random variables with values in subsets of $\mathbb{R}$, $\Gamma_1, \Gamma_2, \dots, \Gamma_N$, respectively. We consider without lost of generality that the random field u has scalar values, i.e., $m = 1$. In practice, one may also be interested in quantities:

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$$y = F(u(\cdot;\boldsymbol{\xi})), \tag{2}$$

that are functions of the solution u of the boundary value problem (1), in addition to the solution itself. In computational fluid dynamics for instance, u may be the pressure field satisfying the compressible Navier–Stokes equations about a fixed profile, and y may be the aerodynamic forces (e.g., drag, lift) exerted by the flow on that profile. In this latter case, the differential operator $\mathscr{L}$ may also depend on time t, and the boundary value problem (1) needs be supplemented with initial conditions. We do not consider that more general situation in the following discussion, for its main features basically extend to time-dependent problems.

Probabilistic Framework

The vector of random parameters $\boldsymbol{\xi}$ is representative of variable geometrical characteristics, boundary conditions, loads, physical or mechanical properties, or combinations of them. It can be discrete, continuous, or a combination of both. In the continuous case, it is understood that its probability distribution $\mathscr{P}_\Xi$ admits a probability density function (PDF) $\boldsymbol{\xi} \mapsto W_\Xi(\boldsymbol{\xi})$ with values in $\mathbb{R}_+ = [0, +\infty]$ such that $\mathscr{P}_\Xi(B) = \int_B W_\Xi(\boldsymbol{\xi}) d\boldsymbol{\xi}$ for any subset B of $\mathbb{R}^N$. In addition, $\mathscr{P}_\Xi(d\boldsymbol{\xi}) = \mathscr{P}_1(d\xi_1) \times \mathscr{P}_2(d\xi_2) \times \cdots \times \mathscr{P}_N(d\xi_N)$ owing to the assumption of mutual independence. In the present setting, it is further assumed that the random parameters are exponentially integrable, that is there exists $\beta > 0$ such that:

$$\int_{\mathbb{R}^N} \mathrm{e}^{\beta\|\boldsymbol{\xi}\|} \mathscr{P}_\Xi(d\boldsymbol{\xi}) < +\infty, \tag{3}$$

where $\|\boldsymbol{\xi}\| = (\sum_{n=1}^N \xi_n^2)^{\frac{1}{2}}$ is the usual Euclidean norm in $\mathbb{R}^N$. Any distribution with compact support satisfies the above condition, for example uniform or beta distributions, as well as the gamma or normal distributions. Together with mutual independence, it ensures that each random variable ξ_n possesses finite moments of all orders, that is $\mathbb{E}\{|\xi_n|^k\} = \int_{\mathbb{R}} |\xi|^k \mathscr{P}_n(d\xi) < +\infty$ for all $k \in \mathbb{N}$. This (uniquely) defines a sequence of univariate orthonormal polynomials $\{\psi_j^{(n)}\}_{j\in\mathbb{N}}$ associated with the probability measure $\mathscr{P}_n$ for all $1 \leq n \leq N$, and a sequence of multivariate orthonormal polynomials $\{\psi_{\mathbf{j}}\}_{\mathbf{j}\in\mathbb{N}^N}$ associated with the probability measure $\mathscr{P}_\Xi$ given by:

$$\psi_{\mathbf{j}}(\boldsymbol{\xi}) = \prod_{n=1}^N \psi_{j_n}^{(n)}(\xi_n), \quad \mathbf{j} = (j_1, j_2, \ldots, j_N) \in \mathbb{N}^N, \tag{4}$$

such that $\{\psi_{\mathbf{j}}(\boldsymbol{\xi})\}_{\mathbf{j}\in\mathbb{N}^N}$ constitutes an orthonormal sequence of random variables in the space $L^2(\Omega, \sigma(\boldsymbol{\xi}), \mathscr{P})$ of second-order random variables defined on the probability space endowed with the σ-algebra $\sigma(\boldsymbol{\xi})$ generated by the random parameters $\boldsymbol{\xi}$; see [1, Theorem 3.6]. Alternatively, the polynomial set $\{\psi_{\mathbf{j}}\}_{\mathbf{j}\in\mathbb{N}^N}$ constitutes an orthonormal basis of the functional space $L^2(\mathbb{R}^N, \sigma_B(\mathbb{R}^N), \mathscr{P}_\Xi(d\boldsymbol{\xi}))$ of square-integrable functions with respect to $\mathscr{P}_\Xi$, where $\sigma_B(\mathbb{R}^N)$ is the Borel σ-algebra on $\mathbb{R}^N$.

Consequently, any random variable u in $L^2(\Omega, \sigma(\boldsymbol{\xi}), \mathscr{P})$ can be expanded in a series of multivariate orthonormal polynomials in the random parameters $\boldsymbol{\xi}$ as:

$$u = \sum_{\mathbf{j}\in\mathbb{N}^N} u_{\mathbf{j}}\psi_{\mathbf{j}}(\boldsymbol{\xi}), \quad u_{\mathbf{j}} = \mathbb{E}\{u\psi_{\mathbf{j}}(\boldsymbol{\xi})\} = \int_{\mathbb{R}^N} u\psi_{\mathbf{j}}(\boldsymbol{\xi})\mathscr{P}_\Xi(d\boldsymbol{\xi}). \tag{5}$$

This is the so-called generalized polynomial chaos expansion. Likewise, the random field $\mathbf{x} \mapsto u(\mathbf{x};\boldsymbol{\xi})$ satisfying Eq. (1) has finite energy from physical considerations, so it belongs to $L^2(\mathbb{R}^N, \sigma_B(\mathbb{R}^N), \mathscr{P}_\Xi(d\boldsymbol{\xi}))$ and can be expanded as:

$$u(\mathbf{x};\boldsymbol{\xi}) = \sum_{\mathbf{j}\in\mathbb{N}^N} u_{\mathbf{j}}(\mathbf{x})\psi_{\mathbf{j}}(\boldsymbol{\xi}), \quad u_{\mathbf{j}}(\mathbf{x}) = \mathbb{E}\{u(\mathbf{x};\boldsymbol{\xi})\psi_{\mathbf{j}}(\boldsymbol{\xi})\} = \int_{\mathbb{R}^N} u(\mathbf{x};\boldsymbol{\xi})\psi_{\mathbf{j}}(\boldsymbol{\xi})\mathscr{P}_\Xi(d\boldsymbol{\xi}). \tag{6}$$

In practical numerical applications, the foregoing expansions are truncated up to a total order p such that $|\mathbf{j}| = j_1 + j_2 + \cdots + j_N \leq p$. Denoting by $\mathbb{P}^p[\cdot]$ the orthogonal projection onto the space of N-variate polynomials of total degree p in $\xi_1, \xi_2, \ldots, \xi_N$, say V_N^p, we seek for an approximate solution $\mathbb{P}^p[u]$ of Eq. (1) in V_N^p as:

$$u(\mathbf{x};\boldsymbol{\xi}) \simeq \mathbb{P}^p[u](\mathbf{x};\boldsymbol{\xi}) = \sum_{|\mathbf{j}|\leq p} u_{\mathbf{j}}(\mathbf{x})\psi_{\mathbf{j}}(\boldsymbol{\xi}),$$

which, by reordering the multi-indices $\mathbf{j}$ such that $|\mathbf{j}| \leq p$, also reads:

$$u(\mathbf{x};\boldsymbol{\xi}) \simeq \mathbb{P}^p[u](\mathbf{x};\boldsymbol{\xi}) = \sum_{j=0}^{P} u_j(\mathbf{x})\psi_j(\boldsymbol{\xi}), \quad P + 1 = \binom{N+p}{p}. \tag{7}$$

From [1, Theorem 2.2], the sequence $\mathbb{P}^p[u]$ converges to u in the mean-square sense in $L^2(\Omega, \sigma(\boldsymbol{\xi}), \mathscr{P})$ provided that the condition (3) is fulfilled, i.e.,

$$\lim_{p\to+\infty} \mathbb{E}\{|u - \mathbb{P}^p[u]|^2\} = 0.$$

Mean-square convergence classically implies convergence in probability, which in turn implies convergence in distribution, the converse being of course untrue in the general case.

Now the deterministic functional coefficients $\mathbf{x} \mapsto u_j(\mathbf{x})$ in the truncated series remain unknown since the random field u is unknown. Weighted versions of Eq. (1) together with the above approximation are considered in order to determine them.

Stochastic Galerkin Method

Similarly to the weak formulation of deterministic problems, one can form the weak form of Eq. (1) and seek an approximate solution $u^p \in V_N^p$ such that:

$$\begin{aligned} \mathbb{E}\{\mathscr{L}(\mathbf{x},\boldsymbol{\xi};u^p)v(\boldsymbol{\xi})\} &= 0 \quad \forall v(\boldsymbol{\xi}) \in V_N^p, \mathbf{x} \in \mathscr{D}, \\ \mathbb{E}\{\mathscr{B}(\mathbf{x},\boldsymbol{\xi};u^p)v(\boldsymbol{\xi})\} &= 0 \quad \forall v(\boldsymbol{\xi}) \in V_N^p, \mathbf{x} \in \partial\mathscr{D}, \end{aligned} \tag{8}$$

where $\mathbb{E}\{\cdot\}$ stands for the mathematical expectation. The resulting system becomes a deterministic one in the physical domain $\mathscr{D}$ for the coefficients $u_j(\mathbf{x})$ and may be solved by standard discretization techniques, e.g., finite elements, finite volumes, finite differences, boundary elements, etc; see [2] and references therein for an extensive presentation of this method.

Stochastic Collocation Method

Alternatively, one may seek an approximate solution formed by interpolation between solutions of Eq. (1) for Q particular choices of the random parameters $\boldsymbol{\xi}$, namely the set $\{\boldsymbol{\xi}^l\}_{1\leq l\leq Q}$ of so-called nodes, such that:

$$\begin{aligned} \mathscr{L}(\mathbf{x},\boldsymbol{\xi}^l;u(\mathbf{x};\boldsymbol{\xi}^l)) &= 0 \quad \forall l = 1,2,\dots,Q, \mathbf{x} \in \mathscr{D}, \\ \mathscr{B}(\mathbf{x},\boldsymbol{\xi}^l;u(\mathbf{x};\boldsymbol{\xi}^l)) &= 0 \quad \forall l = 1,2,\dots,Q, \mathbf{x} \in \partial\mathscr{D}. \end{aligned} \tag{9}$$

Then the approximate solution $\mathbb{I}^Q[u]$ to Eq. (1) reads as the Lagrange interpolation [3–6]:

$$u(\mathbf{x};\boldsymbol{\xi}) \simeq \mathbb{I}^Q[u](\mathbf{x};\boldsymbol{\xi}) = \sum_{l=1}^{Q} u(\mathbf{x};\boldsymbol{\xi}^l)L_l(\boldsymbol{\xi}), \tag{10}$$

where $\{L_l\}_{1\leq l\leq Q}$ is the set of N-variate Lagrange polynomials based on the nodes $\{\boldsymbol{\xi}^l\}_{1\leq l\leq Q}$ chosen so that uniqueness of the interpolation is ensured. Sufficient conditions for such a proper interpolation are outlined in [7]. Choosing the nodes within a quadrature rule $\boldsymbol{\Theta}(N,Q) = \{\boldsymbol{\xi}^l, w^l\}_{1\leq l\leq Q}$ tailored such that $\sum_{l=1}^{Q} w^l f(\boldsymbol{\xi}^l)$ is a good approximation of the N-dimensional integral $\int_{\mathbb{R}^N} f(\boldsymbol{\xi})\mathscr{P}_{\boldsymbol{\Xi}}(d\boldsymbol{\xi}) = \mathbb{E}\{f(\boldsymbol{\xi})\}$ for sufficiently smooth functions f, and the collocation approach may be used to compute an approximate solution $\mathbb{P}_Q^p[u]$ defined by:

$$\begin{aligned}\mathbb{P}^p_Q[u](\mathbf{x};\boldsymbol{\xi}) &= \sum_{j=0}^{P}\left(\sum_{l=1}^{Q} w^l u(\mathbf{x};\boldsymbol{\xi}^l)\psi_j(\boldsymbol{\xi}^l)\right)\psi_j(\boldsymbol{\xi}) \\ &= \sum_{l=1}^{Q} u(\mathbf{x};\boldsymbol{\xi}^l)\left(w^l\sum_{j=0}^{P}\psi_j(\boldsymbol{\xi}^l)\psi_j(\boldsymbol{\xi})\right) \\ &= \sum_{l=1}^{Q} u(\mathbf{x};\boldsymbol{\xi}^l)\tilde{L}_l(\boldsymbol{\xi})\end{aligned} \tag{11}$$

in view of Eq. (7); that is, the quadrature set $\boldsymbol{\Theta}(N,Q)$ is used to evaluate the coefficients $u_j(\mathbf{x})$ in Eq. (7). The latter approach is called pseudo-spectral collocation in [8]. Provided that the quadrature rule $\boldsymbol{\Theta}(N,Q)$ integrates exactly all N-variate polynomials of total order $2p$ and $L_l \in V^p_N$, one has $\tilde{L}_l \equiv L_l$ owing to the orthonormalization of the polynomials $\{\psi_j\}_{0\leq j\leq P}$ which are such that $\mathbb{E}\{\psi_j(\boldsymbol{\xi})\psi_k(\boldsymbol{\xi})\} = \delta_{jk}$, the Kronecker symbol.

Regression

In regression approaches, the $P+1$ expansion coefficients in Eq. (7) are determined on the basis of a set of observations $\{u(\cdot;\boldsymbol{\xi}^l\}_{1\leq l\leq Q}$, obtained by computations or measurements, of the random variable or field u for some particular choices of the random parameters $\boldsymbol{\xi}$, again the set $\{\boldsymbol{\xi}^l\}_{1\leq l\leq Q}$. They consist in solving a weighted least-squares minimization problem:

$$\mathbf{U} = \arg\min_{\hat{\mathbf{U}}\in\mathbb{R}^{P+1}} \frac{1}{2}\left(\mathbf{y}-\boldsymbol{\Phi}\hat{\mathbf{U}}\right)^{\mathsf{T}}\mathbf{W}\left(\mathbf{y}-\boldsymbol{\Phi}\hat{\mathbf{U}}\right), \tag{12}$$

where $\mathbf{y} = (u(\cdot;\boldsymbol{\xi}^1), u(\cdot;\boldsymbol{\xi}^2), \ldots, u(\cdot;\boldsymbol{\xi}^Q))^{\mathsf{T}}$ is the vector of observations, $[\boldsymbol{\Phi}]_{lj} = \psi_j(\boldsymbol{\xi}^l)$ is the so-called $Q\times(P+1)$ measurement matrix, $\mathbf{W}$ is a $Q\times Q$ weighting matrix, and $\mathbf{U} = (u_0, u_1, \ldots, u_P)^{\mathsf{T}}$ is the sought vector of coefficients. This is the approach retained in, e.g., [9, 10], for which numerous methods are available to solve this optimization problem whenever $Q > P$. In [11–14], for example, the choice of the observation set $\{\boldsymbol{\xi}^l\}_{1\leq l\leq Q}$ is also discussed. Alternatively, one may consider the situation whereby $Q\leq P$ and more particularly $Q\ll P$, that is, underdetermined systems. This can be achieved thanks to some recent mathematical results pertaining to the resolution of under-sampled linear systems promoting sparsity of the sought solution, known as compressed sensing or compressive sampling [15, 16]. A recent review of the application of this approach to generalized polynomial chaos expansions is proposed in [17]; see also [18] for an application in the context of the UMRIDA project. The compressed sensing approach consists in reformulating the least-squares minimization problem (12) as a convex minimization problem with some sparsity constraint, namely

$$\mathbf{U} = \arg \min_{\hat{\mathbf{U}} \in \mathbb{R}^{P+1}} \{ \|\hat{\mathbf{U}}\|_1 ; \; \|\mathbf{W}(\mathbf{y} - \boldsymbol{\Phi}\hat{\mathbf{U}})\|_2 \leq \varepsilon \}, \tag{13}$$

for some tolerance $0 \leq \varepsilon \ll 1$ on the polynomial chaos truncation (7). Here the ℓ_m-norm is $\|\mathbf{a}\|_m = (\sum_{j=0}^{P} \left|a_j\right|^m)^{\frac{1}{m}}$ for $m > 0$, and $\|\mathbf{a}\|_0 = \#\{j;\, a_j \neq 0\}$ otherwise. Sparsity means that only a small fraction of the sought coefficients $\mathbf{U}$ are non-negligible. The latter problem is known as basis pursuit denoising (BPDN) [19]. It is uniquely solved thanks to some ad hoc mixing properties of the measurement matrix $\boldsymbol{\Phi}$ which are briefly reviewed in the chapter "Generalized Polynomial Chaos for Non-intrusive Uncertainty Quantification in Computational Fluid Dynamics" of this Book.

Intrusive Versus Non-intrusive Methods: General Comments

The stochastic Galerkin method is **intrusive** in that it may require significant alterations of the existing deterministic codes used for solving numerically the boundary value problem (1). Generally, it also yields coupled equations for the expansion coefficients of its solutions. Hence, new codes need be developed to handle the larger and coupled systems of equations arising from the Galerkin formulation. A recent application of an intrusive approach to the 3D compressible Navier–Stokes equations can be found in Dinescu et al. [20]. The stochastic collocation method and regression methods are **non-intrusive** in that they require only repetitive executions of the existing deterministic codes for carefully selected parameters values. They are the preferred methodologies in CFD, and for their applicability is not affected by the complexity and high nonlinearity of the existing flow solvers so long as they achieve a reasonable accuracy. However, it is to be expected that, for a fixed total order of the polynomial expansion, say, the non-intrusive methods may require the solutions of a much larger number of equations than that of the intrusive Galerkin method, especially for higher dimensions N of the stochastic space [21]. The aliasing error induced by the numerical quadrature rules used in collocation methods may increase with the stochastic dimension alike, indicating that the intrusive Galerkin method may actually offer the most accurate and least demanding solutions in higher dimensions.

A final remark worth to mention is that most of the techniques developed so far, being either intrusive or non-intrusive, have been tested for elliptic and parabolic equations, of which solutions are known to be smooth with respect to space and time. Hyperbolic systems have been much less studied, one important issue being the consideration of the characteristics of the original equations in their Galerkin weak formulation. This issue is addressed in, e.g., [22–24], though.

Overview of the Methodologies Used in UMRIDA

Within UMRIDA, four different non-intrusive methodologies have been implemented. They are described shortly hereafter, with their main advantages and disadvantages. For more details, the reader is referred to the UMRIDA Book.

Sparse Quadrature Sampling Schemes

NUMECA and ONERA both developed a stochastic collocation (SC) method using possibly sparse numerical quadrature rules $\Theta(N, Q)$. The SC expansion of a time-dependent random field $\mathbf{x}, t \mapsto u(\mathbf{x}, t; \boldsymbol{\xi})$ reads:

$$u(\mathbf{x}, t; \boldsymbol{\xi}) \simeq \mathbb{I}^Q[u](\mathbf{x}, t; \boldsymbol{\xi}) = \sum_{l=1}^{Q} u(\mathbf{x}, t; \boldsymbol{\xi}^l) L_l(\boldsymbol{\xi}), \tag{14}$$

where the expansion coefficients are the random field evaluated at the nodes $\{\boldsymbol{\xi}^l\}_{1 \le l \le Q}$. In stochastic dimension one ($N = 1$), the nodes ξ^l are chosen as the numerical quadrature points associated with the PDF $\xi \mapsto W_1(\xi)$ of the single random parameter ξ. The moments are then calculated using numerical quadrature, e.g., for the mean:

$$\begin{aligned} \mathbb{E}\{u\}(\mathbf{x}, t) &= \int_{\Gamma_1} u(\mathbf{x}, t; \xi) W_1(\xi) d\xi \\ &\simeq \sum_{l=1}^{Q} w^l u(\mathbf{x}, t; \xi^l), \end{aligned} \tag{15}$$

and similar expressions for the higher-order moments. In higher stochastic dimensions, the numerical quadrature formulas involve tensor products of the one-dimensional quadrature nodes. Such a tensorization quickly becomes very expensive in terms of the number of sampling nodes—the so-called curse of dimensionality. NUMECA therefore used a sparse quadrature sampling scheme, based on Smolyak's algorithm [25]. It significantly reduces the number of samples at the cost of ignoring some higher-order coupling terms in the expansion. Since the expansion coefficients of these terms can supposedly be small, this has only a small impact on the accuracy of the calculated moments. The basic methodology is for scalar random data but by repeating it, it can also be used for random fields.

Adaptive Sparse Polynomial Chaos

This methodology is used by AGI and ESTECO. It is based on the regression approach developed by Blatman and Sudret [26–28]. The basic idea of this approach is that in the classical polynomial chaos (PC) expansion of a time-dependent random field:

$$u(\mathbf{x}, t; \boldsymbol{\xi}) \simeq \mathbb{P}^P[u](\mathbf{x}, t; \boldsymbol{\xi}) = \sum_{j=0}^{P} u_j(\mathbf{x}, t)\psi_j(\boldsymbol{\xi}), \tag{16}$$

the expansion coefficients $u_j(\mathbf{x}, t)$ of many of the higher-order terms are negligible. An iterative procedure is developed to find out which are the important terms to be retained. The procedure is based on least angle regression (LAR, as opposed to numerical quadrature) and is quite efficient. Compared to the classical regression approach by least-squares minimization, where oversampling is used with typically $Q = 2P$ samples, under-sampling is used here. The size of the experimental design (ED) is difficult to select a priori but a sequential one can be chosen. A leave-one-out procedure is used to test the quality of the proposed expansion. Apart from the size of the ED, the choice of the location of the sampling nodes is also important. Depending on the problem at hand, the solution might be more or less sensitive to this choice. The basic methodology is for scalar random data but a procedure has been proposed for random fields as well based on proper orthogonal decomposition (POD) in [29].

Compressive Sampling

This methodology is used by ONERA. As in the adaptive sparse PC expansion of the foregoing section, it exploits the fact that many terms in the PC expansion of Eq. (16) are negligible. The leading expansion coefficients are determined by regression from an underdetermined system with the following additional condition:

$$\|\mathbf{U}\|_1 = \sum_{j=0}^{P} \left|u_j(\mathbf{x}, t)\right| \quad \text{is minimal.} \tag{17}$$

This latter condition expresses the sparsity of the expansion; see Eq. (13). Different software packages are available to solve Eq. (13), such as `SPGL1` [30] or `CVXOPT` [31]. The obtained result is comparable to that of the adaptive sparse PC expansion but there is of course some dependency on the used software. However, the ℓ_1-minimization procedure has the great advantage of being non adaptive, as it identifies both the amplitude and the rank of the leading coefficients in a single run. This is a much desirable feature for practical industrial applications with complex configurations. The size of the ED and the location of the sampling points can also have an effect on the quality of the solution. The basic methodology is for scalar stochastic data but by repeating it, it can also be used for stochastic fields.

Reduced Basis Method

This methodology is developed by VUB. Instead of the classical PC expansion of Eq. (16) with the polynomials $\{\psi_j(\boldsymbol{\xi})\}_{0\leq j\leq P}$ as the basis, an expansion in a reduced basis is used:

$$u(\mathbf{x}, t; \boldsymbol{\xi}) \simeq \sum_{j=0}^{m} \hat{u}_j(\mathbf{x}, t) z_j(\boldsymbol{\xi}), \tag{18}$$

where the new basis is formed by $\{z_j(\boldsymbol{\xi})\}_{1\leq j\leq m}$ and $m \ll P$ so that less expansion coefficients have to be determined. The new basis can be determined via a Karhunen–Loève expansion, also known as POD. This requires however the knowledge of the covariance of the random field $u(\mathbf{x}, t; \boldsymbol{\xi})$. Assuming that this covariance (in stochastic space) is largely independent from the resolution in real space, it can be determined by a PC analysis on a coarse grid. Once the reduced basis is found, the expansion

Table 1 Comparison of different PC methods used in the UMRIDA project

Method	Idea	Methodology	Technique	Application
Sparse quadrature PC	Cheaper numerical quadrature	Sparse instead of tensorial quadrature rule	Numerical quadrature	Scalar data extendable to fields by repeating procedure
Sparse PC	Only most important terms in PC decomposition are kept	Different expansions tried in a systematic and iterative way	Regression LAR or stepwise	Scalar data extendable to fields using POD
Compressive sampling	Only most important terms in PC decomposition are kept	Found in one shot	Regression with constraint	Scalar data extendable to fields by repeating procedure
Reduced basis	Find optimal PC decomposition based on covariance and POD	PC on coarse grid. All above techniques can be used to reduce CPU on coarse mesh	Regression on fine grid	Fields >1 field: use covariance of all fields scalar data: use reduced basis of the field data (scalar data depends on field data)

coefficients are computed with (overdetermined) regression, typically using $Q = 2m$ samples. The size of the reduced basis depends on the number of POD modes taken into account. Normally, only few modes suffice, but if the correlation length is small, more modes are needed for an accurate representation. Nevertheless, $m \ll P$ so that significantly less samples are needed on the fine grid compared to a "full" PC expansion. It is also to be noted that for the coarse grid PC analysis, any of the other methods discussed here—e.g., compressive sampling, sparse adaptive PC expansion—can be used to further increase the efficiency. As the methodology is based on the covariance, it is directly applicable to random fields solely. In case of different random fields, the covariance of all the fields is to be used in principle. However, tests with the RAE2822 basic case considered in the UMRIDA project showed that, using only one random field, e.g., the pressure field, the resulting reduced basis is very similar to that obtained using all random fields. For scalar data, which are usually derived from (a) random field(s), the reduced basis derived from these field(s) can be used. The different methodologies are summarized in Table 1.

Comparison of the Different Methodologies in Terms of Efficiency

We remind here that the total order of the polynomial expansion (6) is denoted by p. If the stochastic dimension is $N = 1$, then the Lagrange polynomials in Eq. (14) are of order p. Hence, $p + 1$ quadrature points $\{\xi^l\}_{1\leq l\leq p+1}$ are needed. Since $u(\mathbf{x}, t; \xi^l)$ in Eq. (14) is the value of $u(\mathbf{x}, t)$ at the quadrature point ξ^l, this implies that $p + 1$ simulations are needed. If the stochastic dimension is $N > 1$, this becomes $(p + 1)^N$ in the case of a fully tensorized scheme based on $p + 1$ nodes in each dimension. Using a sparse quadrature rule instead, different levels k can be considered, where k varies from $k = 1$ up to $k = p + 1$. The number of quadrature points, and hence the number of simulations, for a sparse quadrature of level k and $N \gg 1$ can be approximated as:

$$Q \simeq \frac{(2N)^k}{k!}. \tag{19}$$

If we set to unity the cost of a single simulation, this is also the total cost. It should be noted that, with the full tensorial scheme, polynomials up to a total order $N(2p + 1)$ (i.e., a product of N polynomials each of order $2p + 1$ in each dimension) are integrated exactly, whereas, with the sparse scheme of level k based on an underlying one-dimensional Gauss rule with $p + 1$ nodes, only polynomials up to total order $2k - 1$ are integrated exactly, i.e., maximal order $2p + 1$ for the highest level $k = p + 1$.

In the reduced basis method implemented by VUB, a full PC expansion constructed by regression is first considered on a **coarse** grid. The total number of samples required for the regression is $Q = 2(P + 1)$, where $P + 1$ is again the number of terms in the expansion of Eq. (16). In stochastic dimension N, $P + 1 = \binom{N+p}{p}$.

The total cost is therefore $2c \times \binom{N+p}{p}$ where c takes into account the coarseness of the grid; that is, c is the ratio of coarse to fine grid size. Next, regression is used on the fine grid. Assuming the reduced basis has dimension m, this requires $2m$ fine grid samples, with a cost $2m$. Based on the RAE2822 and ROTOR37 test cases defined in the UMRIDA project where respectively $N = 10$ and $N = 21$ uncertain parameters were considered, a reduced basis of size $m \simeq 20$ can be expected, whereas for the value of the coarse to fine grid size $c = \frac{1}{8}$ is found acceptable. The total cost can therefore be estimated as $\frac{1}{4} \times \binom{N+p}{p} + 40$.

In the adaptive sparse PC expansion and compressive sampling approaches, the number of samples Q is not directly related to p and N. However, as an underdetermined system of equations is considered for the regression analysis, Q is (much) smaller than $P + 1$. ESTECO, for instance, used $Q = 200$ samples for the RAE2822 test case with $N = 13$ uncertainties, going up to the PC order $p > 10$. A similar number can be derived from the literature. Blatman and Sudret [28] mention speedups of 3.3–10.2 compared to full PC expansions of order 2, when the stochastic dimension varies from $N = 30$–70. This leads to an equivalent of $Q = 300$–500 samples. In compressive sampling, ONERA used $Q = 80$ samples for the RAE2822 test case with $N = 3$ uncertainties. As a rule of thumb, the number of samples to be used in compressive sampling is typically four times the sparsity $S \equiv \|\mathbf{U}\|_0$ of the sought solution, that is the number of non-negligible coefficients: $Q \geq 4S$. In Hampton and Doostan [32], errors of less than 1% in the mean (compared to full PC expansion) are obtained with about 200–300 samples for an example with $N = 20$. A choice of 100 samples for $N = 5$, 180 samples for $N = 10$ and 250 samples for $N = 20$ therefore seems appropriate for both methods.

Based on the reasoning above, which admittedly gives only rough cost estimates, the costs of the different methods are compared in Tables 2 through 4 below for $N = 5$, $N = 10$ and $N = 20$, respectively. The first two rows show the number of samples of full PC with respectively regression (assuming $2P$ samples) and a ten-

Table 2 Cost of different PC methods used in UMRIDA for $N = 5$ and varying total order p

Method $N = 5$	$p = 1$	$p = 2$	$p = 3$	$p = 4$
Full PC regression	12	42	112	252
Tensorial quadrature PC	32	243	1024	3125
Sparse quadrature PC	50 (11)	166 (71)	417 (341)	833 (1341)
Adaptive sparse PC	100	100	100	100
Compressive sampling	100	100	100	100
Reduced basis	42	45	54	72

Table 3 Cost of different PC methods used in UMRIDA for $N = 10$ and varying total order p

Method $N = 10$	$p = 1$	$p = 2$	$p = 3$	$p = 4$
Full PC regression	22	132	572	2002
Tensorial quadrature PC	1024	59,049	1,048,576	9,765,625
Sparse quadrature PC	200 (21)	1333 (241)	6666 (1981)	26,666 (12981)
Adaptive sparse PC	180	180	180	180
Compressive sampling	180	180	180	180
Reduced basis	43	47	112	290

Table 4 Cost of different PC methods used in UMRIDA for $N = 20$ and varying total order p

Method $N = 20$	$p = 1$	$p = 2$	$p = 3$	$p = 4$
Full PC regression	42	462	3542	21252
Tensorial quadrature PC	1,048,576	3.486E9	1.099E12	95.367E12
Sparse quadrature PC	800	10,666	106,666	853,333
Adaptive sparse PC	250	250	250	250
Compressive sampling	250	250	250	250
Reduced basis	45	98	483	2697

sorized numerical quadrature with $(p + 1)^N$ samples. For the sparse quadrature sampling and reduced basis approaches, PC total orders up to $p = 4$ are considered. For sparse quadrature sampling, the given cost is for the scheme with the highest level k. According to the results obtained by NUMECA, this highest level is not always required. For the ROTOR37 case, for instance, NUMECA reports differences of only 0.02% and 3%, respectively, on the mean and variance if the level $k = 2$ is used instead of the level $k = 3$. The cost of sparse quadrature sampling using lower levels is found in the columns more to the left. That is, for a PC order $p = 4$ the number of samples required for $k = 4$ is found in the column $p = 4$, those of $k = 3$ in column $p = 3$, those of $k = 2$ in column $p = 2$ etc. (Tables 2 through 4).

It is important to stress once more that the tables above are based on very rough cost estimates. Nevertheless, some conclusions can be drawn. Assuming that, in practice, at least a PC total order of 3 is needed for accurate results, we focus on the last two columns in the tables.

- The adaptive sparse PC and compressive sampling seem the most efficient methods, especially when going to high-stochastic dimensions. Whereas for $N = 5$ the reduced basis is most efficient, this is not the case anymore for $N = 10$ and $N = 20$;
- The sparse quadrature sampling, though much more efficient than the fully tensorized quadrature sampling, seems less efficient than the other methods. When comparing it to the full PC with regression, the sparse quadrature sampling is not competitive for $N = 5$. At the higher stochastic dimensions considered, $N = 10$ and $N = 20$, it can compete with the other methods only if the lowest level is used ($p = 1$). It is stressed that the number of samples mentioned is based on a general formula which gives a rough estimate only. Depending on the numerical quadrature formula used (e.g., with nested or non-nested stencils), the number of samples might be lower. This is illustrated in Tables 2 and 3 for $N = 5$ and $N = 10$ where the number of samples for the sparse numerical formulas used by NUMECA is mentioned in between brackets. In those cases, the sparse quadrature sampling method is competitive with the other methods for the higher stochastic dimensions $N = 5$ and $N = 10$, if one restricts it to the level $k = 2$;
- Looking at the effect of increasing the stochastic dimension N, the efficiency of adaptive sparse PC and compressive sampling increases with it. This is also clearly found in Blatman and Sudret [28], who mention speedups of 3.3–10.2 compared to full PC of order 2, when the stochastic dimension varies from $N = 30$–70. Similarly for the reduced basis, the speedups compared to full PC with regression of the same order p are respectively 3.5, 6.9, and 7.8 (for $p = 4$ and $N = 5, 10$, and 20, respectively). If sparse quadrature sampling is compared to full PC with regression and $p = 4$, there is only a speedup if the formulas of level $k = 1$ or $k = 2$ are used. For level 2, the speedups for $N = 5, 10$, and 20 are respectively 1.5, 1.5, and 2.0—based on the number of samples from the approximate formula (19). With the number of samples used by NUMECA, this becomes 3.5 and 8.3, respectively, for $N = 5$ and 10. We can therefore conclude that, compared to standard PC methods, all considered methods become more efficient with increasing stochastic dimension. Therefore, they allow to partially mitigate the effect of the "curse of dimensionality."

Comparison of Computing Times on UMRIDA Test Case ROTOR37

The NASA ROTOR37 is an axial flow transonic compressor designed and experimentally tested by Reid and Moore [33]. In the UMRIDA project, the working point corresponding to a rotational speed of Ω =17,188 rpm and an outlet static pressure of p_{out} =11,000 Pa is chosen. This corresponds to a mass flow which is 98% of the mass flow at choking conditions.

Only two partners contributed to this test case, using PC methodology: NUMECA with sparse quadrature sampling, and VUB with the reduced basis method.

Table 5 CPU cost in h on 100 cores for the ROTOR37 test case on the actual mesh and corrected for mesh size of 1M cells

Partner	Method	N	Mesh (Mcells)	CPU (h/100 cores)	CPU 1M mesh (h/100 cores)
NUMECA	Sparse quad. $k = 1$	10	2.8	0.56	0.20
	Sparse quad. $k = 2$	10	2.8	6.43	2.29
VUB	Reduced basis	12	0.77	0.23	0.29
	Reduced basis	21	0.77	0.66	0.86

Uncertainties are imposed by both partners on the following operational parameters: the inlet total pressure profile (described with one uncertainty, i.e., assumed fully correlated) and the outlet static pressure. In addition, geometrical uncertainties are introduced on the tip clearance and the blade parametrization: leading and trailing edge angles at different spanwise locations as well as blade half thickness parameters (VUB) and leading edge radius (NUMECA). In total, VUB considered two cases with respectively $N = 12$ and 21 uncertainties, whereas NUMECA used $N = 10$ uncertainties. Both partners used the same simulation code, i.e., the FINE/Turbo solver of NUMECA with RANS and Spalart–Allmaras turbulence model and central scheme with artificial dissipation and multigrid algorithms. NUMECA used a grid with 2.8 million points, VUB with 0.77 million. The PC total order considered was $p = 2$.

Table 5 compares the CPU times. The numbers given are the CPU hours on 100 cores. In the last column, corrected numbers are given, taking into account the difference in grid size: they are rescaled for a grid of 1 million points, hereby assuming the computational cost is linear with grid size (as should be the case when using multigrids). One of the milestones of the UMRIDA project was the ability to handle a large number of uncertainties ($N > 10$), including geometrical uncertainties, in a turnaround time of the order of 10 h on a cluster of 100 cores. This milestone is therefore clearly achieved with the present methodologies. Comparing the corrected times for NUMECA ($N = 10$) with those of VUB ($N = 12$) the estimated results of Table 3 are confirmed, if one takes into account that the actual size of the reduced basis was only $m = 7$ (instead of 20 anticipated in Table 3). According to Table 3, the reduced basis method should be 2.2 times more expensive than the level 1 sparse quadrature method of NUMECA, and 5.1 times more efficient than NUMECA's level 2 method. The actual numbers (from Table 5) are respectively 1.5 times more expensive and 7.9 more efficient, which is thus explained by the smaller reduced basis.

The level 1 results of NUMECA provide only an approximation of the mean and the variance and do not allow for an accurate estimation of the PDF, in contrast to the level 2 results and the reduced basis results. Taking this into account, the reduced basis method seems more efficient, at least for this application. Looking at the VUB results, the effect of the stochastic dimension can also be evaluated. In the case $N = 21$, the number of POD modes considered (i.e., the size of the reduced basis) was such that the sum of the corresponding eigenvalues (in absolute value) was 99.99% of the total sum. In the case $N = 10$, this was only 99.9%. This resulted in a reduced basis of size $m = 21$ for $N = 21$ and $m = 7$ for $N = 10$. If we correct for this effect and assume the same basis size of $m = 21$ for $N = 10$, the CPU time would be 0.38 (instead of 0.23) compared to 0.66 for $N = 21$. This is more or less a linear increase with the number of uncertainties and not an exponential one as observed in classical methods, due to the "curse of dimensionality."

Conclusions

The different methodologies, used in UMRIDA to deal with the "curse of dimensionality," are compared for their efficiency. Based on rough estimates of the number of required simulations, it is found that all methods allow to mitigate the "curse of dimensionality." For the sparse quadrature method and the reduced basis method, this is also confirmed by simulations of the 3D compressible Navier–Stokes test case ROTOR37 with a stochastic dimension of at least $N = 10$. It is also confirmed that the UMRIDA milestone, of a turnaround time of the order of 10 h on a cluster of 100 cores for a case $N = 10$, is clearly achieved. In the ROTOR37 test case, the required simulation time is at least an order of magnitude less.

References

1. Ernst, O.G., Mugler, A., Starkloff, H.J., Ullmann, E.: On the convergence of generalized polynomial chaos expansions. ESAIM. Math. Model. Numer. Anal. **46**(2), 317–339 (2012)
2. Le Maître, O., Knio, O.: Spectral Methods for Uncertainty Quantification: With Applications to Computational Fluid Dynamics. Springer, Dordrecht (2010)
3. Mathelin, L., Hussaini, M., Zang, T.: Stochastic approaches to uncertainty quantification in CFD simulations. Numer. Algorithms **38**(1), 209–236 (2005)
4. Xiu, D., Hesthaven, J.S.: High-order collocation methods for differential equations with random inputs. SIAM J. Sci. Comput. **27**(3), 1118–1139 (2005)
5. Babuška, I., Nobile, F., Tempone, R.: A stochastic collocation method for elliptic partial differential equations with random input data. SIAM J. Numer. Anal. **45**(3), 1005–1034 (2007)
6. Ganapathysubramanian, B., Zabaras, N.: Sparse grid collocation schemes for stochastic natural convection problems. J. Comput. Phys. **225**(1), 652–685 (2007)
7. Chung, K., Yao, T.: On lattices admitting unique Lagrange interpolations. SIAM J. Numer. Anal. **14**(4), 735–743 (1977)
8. Xiu, D.: Efficient collocational approach for parametric uncertainty analysis. Commun. Comput. Phys. **2**(2), 293–309 (2007)
9. Berveiller, M., Sudret, B., Lemaire, M.: Stochastic finite element: a non-intrusive approach by regression. Rev. Européenne de Mécanique Numérique **15**(1–3), 81–92 (2006)
10. Hosder, S., Walters, R.W., Balch, M.: Point-collocation nonintrusive polynomial chaos method for stochastic computational fluid dynamics. AIAA J. **48**(12), 2721–2730 (2010)
11. Gao, Z., Zhou, T.: On the choice of design points for least-square polynomial approximations with application to uncertainty quantification. Commun. Comput. Phys. **16**(2), 365–381 (2014)

12. Arnst, M., Ponthot, J.P.: An overview of nonintrusive characterization, propagation, and sensitivity analysis of uncertainties in computational mechanics. Int. J. Uncertainty Quantif. **4**(5), 387–421 (2014)
13. Hosder, S., Walters, R.W., Balch, M.: Efficient sampling for non-intrusive polynomial chaos applications with multiple uncertain input variables. In: 48th AIAA/ASME/ASCE/AHS/ASC Structures, Structural Dynamics, and Materials Conference, AIAA paper 2007-1939, Honolulu HI, 23–26 April 2007
14. Raisee, M., Kumar, D., Lacor, C.: A non-intrusive model reduction approach for polynomial chaos expansion using proper orthogonal decomposition. Int. J. Numer. Methods Eng. **103**(4), 293–312 (2015)
15. Candès, E.J., Romberg, J.K., Tao, T.: Stable signal recovery from incomplete and inaccurate measurements. Commun. Pure Appl. Math. **59**(8), 1207–1223 (2006)
16. Donoho, D.L.: Compressed sensing. IEEE Trans. Inf. Theory **52**(4), 1289–1306 (2006)
17. Hampton, J., Doostan, A.: Compressive sampling methods for sparse polynomial chaos expansions. In Ghanem, R.G., Higdon, D., Owhadi, H., (eds.) Handbook of Uncertainty Quantification. Springer, Cham, 29 p (2016)
18. Savin, É., Resmini, A., Peter, J.: Sparse polynomial surrogates for aerodynamic computations with random inputs. In: 18th AIAA Non-Deterministic Approaches Conference, AIAA paper 2016-0433, San Diego CA, 4–8 Jan 2016
19. Chen, S.C., Donoho, D.L., Saunders, M.A.: Atomic decomposition by basis pursuit. SIAM J. Sci. Comput. **20**(1), 33–61 (1998)
20. Dinescu, C., Smirnov, S., Hirsch, C., Lacor, C.: Assessment of intrusive and non-intrusive non-deterministic CFD methodologies based on polynomial chaos expansions. Int. J. Eng. Syst. Model. Simul. **2**(1–2), 87–98 (2010)
21. Xiu, D.: Fast numerical methods for stochastic computations: A review. Commun. Comput. Phys. **5**(2–4), 242–272 (2009)
22. Gottlieb, D., Xiu, D.: Galerkin method for wave equations with uncertain coefficients. Commun. Comput. Phys. **3**(2), 505–518 (2008)
23. Poëtte, G., Després, B., Lucor, D.: Uncertainty quantification for systems of conservation laws. J. Comput. Phys. **228**(7), 2443–2467 (2009)
24. Tryoen, J., Le Maître, O., Ndjinga, M., Ern, A.: Intrusive Galerkin methods with unwinding for uncertain nonlinear hyperbolic systems. J. Comput. Phys. **229**(18), 6485–6511 (2010)
25. Smolyak, S.: Quadrature and interpolation formulas for tensor products of certain classes of functions. Sov. Math. Dokl. **4**, 240–243 (1963)
26. Blatman, G., Sudret, B.: Sparse polynomial chaos expansions and adaptive stochastic finite elements using a regression approach. C. R. Méc. **336**(6), 518–523 (2008)
27. Blatman, G., Sudret, B.: An adaptive algorithm to build sparse polynomial chaos expansions for stochastic finite element analysis. Probab. Eng. Mech. **25**(2), 183–197 (2010)
28. Blatman, G., Sudret, B.: Adaptive sparse polynomial chaos expansion based on least angle regression. J. Comput. Phys. **230**(6), 2345–2367 (2011)
29. Blatman, G., Sudret, B.: Sparse polynomial chaos expansions of vector-valued response quantities. In: Proceedings 11th International Conference on Structural Safety and Reliability (ICOSSAR13) (2013)
30. van den Berg, E., Friedlander, M.P.: `SPGL1`: A Solver For Large-scale Sparse Reconstruction (June 2007)
31. Andersen, M., Dahl, J., Vandenberghe, L.: `CVXOPT`: Python Software For Convex Optimization (2016)
32. Hampton, J.: Doostan: compressive sampling of polynomial chaos expansions: convergence analysis and sampling strategies. J. Comput. Phys. **280**, 363–386 (2015)
33. Reid, L., Moore, R.: Performance of single-stage axial-flow transonic compressor with rotor and stator aspect ratios of 1.19 and 1.26, respectively, and with design pressure ratio of 1.82. NASA Technical Paper 1338 (1978)